# The Missing Local Field: A Live $f_{\mathrm{xc}}[\rho(\mathbf{r})]$ Convolution Lets a From-Scratch Screened Electron–Phonon Vertex Cross the Bare Baseline

Seven rounds of a hexa-native screened-vertex search, and the round that reversed a "bare-is-best" closed-negative ruling

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**Abstract**

A hexa-native first-principles engine (QFORGE) computes the electron–phonon coupling $\lambda$ of high-pressure hydrides end-to-end — SCF, DFPT phonons, the el–ph vertex, and the Allen–Dynes / Eliashberg critical temperature — with the aim of removing the external Quantum ESPRESSO (QE) dependence. The hardest sub-problem is the *screened* el–ph vertex: replacing the bare deformation potential $\partial V_{\mathrm{bare}}$ with the self-consistent $\partial V_{\mathrm{scf}}$ that QE's $|g|^2$ implicitly carries. We pre-registered a falsifiable hypothesis: *a from-scratch screened vertex, built channel-by-channel, will eventually enhance $\lambda$ past the bare baseline toward the QE answer key.* For six rounds the hypothesis appeared *falsified*: every screening channel (kernel, operator, normalization, self-consistent Woodbury–Dyson vertex, screened phonon) left $CaH_6$ $\lambda$ at $\sim$3.0–3.1 ($\sim$30% *under* QE 4.376), strictly worse than the bare vertex's own 4.137 (5.47% under QE), and the engine status was provisionally ruled a *closed-negative, bare-is-best* result. Round 7 reverses that ruling. We trace the residual to a single *structurally dead* channel — the spatially-varying local-field exchange–correlation kernel $f_{\mathrm{xc}}[\rho(\mathbf{r})]$, which was never *called* in the production Woodbury vertex (folds$=0$, $\|f_{\mathrm{xc}}\cdot\Delta\rho\| = 0$). Wiring a live $f_{\mathrm{xc}}[\rho(\mathbf{r})]\cdot\Delta\rho(\mathbf{r})$ convolution on a power-of-two FFT–Poisson real-space grid into the screened vertex (folds$=24$, ENGAGED) raises $CaH_6$ $\lambda$ to 4.1518 — the **first** of seven rounds to *cross* the bare baseline (+0.0153 above 4.137), and the first to beat the bare vertex's own QE-distance (rel-$\varepsilon = 5.12\%$ vs. bare 5.47%). The local field was the missing enhancement physics, exactly as the round-5/6 hypothesis predicted. We are honest about what was *not* achieved: 5.12% does *not* meet the $\leq$1% production-migration gate; the residual is the LDA-vs-QE exchange–correlation functional difference (QFORGE screens with an LDA-x + PW92-c ALDA kernel; QE's $|g|^2$ carries the full $\varepsilon^{-1}$), and a round 8 testing a GGA $f_{\mathrm{xc}}$ is in progress. The from-scratch screened engine is therefore *not closed* — it now demonstrably enhances — but it is not yet gate-grade; the validated hybrid route (QE $|g|^2 \to$ QFORGE assembler, rel-$\varepsilon = 1.65 \times 10^{-7}$) remains the production path for gate-grade $T_c$.

## 1 Hypothesis (pre-registered falsifier)

The QFORGE engine must, to fully replace QE, generate the el–phonon matrix elements $|g(\mathbf{k},\mathbf{q},\nu)|^2$ from scratch. The non-trivial physics is *screening*: the bare deformation potential $\partial V_{\mathrm{bare}}$ that an atomic displacement produces is screened by the electron gas to a self-consistent $\partial V_{\mathrm{scf}}$, and it is $\partial V_{\mathrm{scf}}$ — not $\partial V_{\mathrm{bare}}$ — that QE's $|g|^2$ carries. We pre-registered the following falsifiable hypothesis **(H)**:

> **(H)** A from-scratch self-consistent screened vertex, assembled channel-by-channel, will *enhance* the $CaH_6$ coupling $\lambda$ past the bare-vertex baseline ($\lambda_{\mathrm{bare}} = 4.137$) and

toward the QE answer key ($\lambda_{\mathrm{QE}} = 4.376$). *Falsifier:* if, after every screening channel is built and measured, no screened configuration crosses $\lambda_{\mathrm{bare}}$, then screening as implemented *attenuates* rather than enhances, and the bare vertex is the best from-scratch estimate.

The falsifier is sharp: a single number ($\lambda_{\mathrm{bare}} = 4.137$) that the screened vertex must exceed. The production-migration gate is sharper still: rel-$\varepsilon = |\lambda_{\mathrm{QFORGE}} - \lambda_{\mathrm{QE}}|/\lambda_{\mathrm{QE}} \leq 1\%$ [1, 2].

## 2 Method

**System and convergence.** The anchor is $CaH_6$ in the $Im\bar{3}m$ clathrate structure at $\sim$ 150 GPa, a 7-atom cell with a QE textbook-proof reference $\lambda_{\mathrm{QE}} = 4.376$ [3, 4]. All QFORGE runs use `ecutwfc` = 80 Ry, $n(\mathrm{PW}) = 645$ plane waves (full ecut shell, `npw_cap` = 0), a $2^3$ Monkhorst–Pack $q$-mesh, Gaussian smearing $\sigma = 0.02$ Ha, and an LDA-x + PW92-c exchange–correlation functional. The bare baseline $\lambda_{\mathrm{bare}} = 4.13647$ is the converged full-basis bare-vertex run (`qforge-lane1-basis-sweep`, rel-$\varepsilon = 5.47\%$).

**Screened-vertex channels (rounds 2–6).** The screened vertex routes through a Woodbury low-rank Dyson solve `qdvs_solve_exact` for the self-consistent $\partial V_{\mathrm{scf}}$ column [5, 6]. Across rounds we built and measured each channel independently: the ALDA-floor $f_{\mathrm{xc}}$ kernel (r2), a static-Lindhard $\varepsilon(q)$ operator with $N_{\mathrm{tot}}^2/\Omega$ density normalization (r3/r4), the exact Woodbury self-consistent vertex with RPA Adler–Wiser $\chi_0$ (r5), and a screened phonon force-constant (r6). Each round's verdict is archived verbatim under `.verdicts/qforge-cah6-*/`.

**The dead channel (root cause).** The Woodbury production vertex carries only a *uniform-gas scalar* $f_{\mathrm{xc}}$ head, $K(\mathbf{G}) = \langle v_c \rangle + \bar{f}_{\mathrm{xc}}$. The *spatially-varying* local field $f_{\mathrm{xc}}[\rho(\mathbf{r})] \cdot \Delta\rho(\mathbf{r})$ — the beyond-$\bar{\rho}$ term that QE's $\varepsilon^{-1}|g|^2$ carries — was implemented (`qpwfft_dvscr_from_dpsi`) but **never called** in the production vertex. The round-6 witnesses confirm this: folds = 0, local-ALDA-folds= 0, $\|f_{\mathrm{xc}} \cdot \Delta\rho\| = 0 \Rightarrow$ *diagonal fallback*.

**The round-7 fix.** We diagnosed — and corrected a prior misdiagnosis: the power-of-two FFT–Poisson padding was *never* the blocker (it folds non-zero at any $n$; the production stage already padded $n$=645 $\rightarrow$ grid $32^3$). The real blocker was that the live $f_{\mathrm{xc}}$ convolution was simply absent from the Woodbury column. We added a new routine `qpwfft_fxc_localfield_from_dpsi` returning *only* the local-field $f_{\mathrm{xc}}[\rho(\mathbf{r})] \cdot \Delta\rho(\mathbf{r})$ term (Hartree excluded — Woodbury carries the Coulomb term, no double-count), folded on the power-of-two padded real-space FFT grid at the full $n$=645 basis, and wired it into the screened vertex: after the Woodbury $\partial V_{\mathrm{scf}}$ column, a Sternheimer solve yields $\Delta\psi$, the local-field $f_{\mathrm{xc}}$ is folded, and $\Delta V_{f_{\mathrm{xc}}}(\mathbf{G}) \cdot \psi(\mathbf{G})$ is added to the $\Delta V_{\mathrm{scr}}|\psi\rangle$ vertex column.

## 3 Measurement (verbatim, d6 — not tuned)

The round-7 run is reported verbatim from `.verdicts/qforge-cah6-fxc-localfield-r7/VERDICT.md`; no value was tuned toward the answer key.

```
convergence : ecutwfc=80 Ry, n(PW)=645, q-mesh=2^3 MP, sigma=0.02 Ha,
              xc=LDA-x + PW92-c
vertex      : SCREENED dV_scf (Woodbury Dyson)
              + LIVE local-field f_xc[rho(r)] convolution
f_xc-live   : POW2-FFT-POISSON grid=32x32x32  folds=24  local-ALDA-folds=24
              xc-pts=27648  -> "ENGAGED -- f_xc[rho(r)]"   (R6: folds=0,
                 DEAD)
```

```
vertex ratio: ||dV_scf|| / ||dV_bare|| = 0.984635
BARE baseline lambda = 4.13647   (rel-eps = 5.47%)
QFORGE lambda (R7)   = 4.1518
QE answer-key lambda = 4.376
rel-eps              = 0.0512333  (5.12333 %)
Delta-lambda vs bare = +0.0153329  <- CROSSES BARE (first in 7 rounds)
QFORGE omega_log     = 1379.56 K
QFORGE Tc (Allen-Dynes) = 386.65 K
QFORGE Tc (Eliashberg)  = 415.75 K
GATE: NOT MET -- rel-eps = 5.12% > 1%
```

The $f_{xc}$-live witnesses (folds= 24, local-ALDA-folds= 24, xc-pts= 27648) confirm the previously dead channel is engaged: a genuine RPA+ALDA local-field convolution, $f_{xc}[\rho(\mathbf{r})]$.

## 4 Finding

**The $\lambda$ trajectory (the central result).** Table 1 and Figure 1 give the full seven-round trajectory. Round 7 is the **first** screened result to (a) exceed the bare baseline $\lambda_{\mathrm{bare}} = 4.137$ and (b) beat the bare vertex's own QE-distance (5.12% < 5.47%).

**Table 1:** $CaH_6$ screened-vertex $\lambda$ trajectory across seven rounds. Round 7 is the first to cross the bare baseline. All values verbatim from the per-round verdict directories; rel-$\varepsilon$ is vs. QE 4.376.

| Round | Channel added | $\lambda$ | rel-$\varepsilon$ | Verdict dir |
|---|---|---|---|---|
| bare | screening OFF (mode a) | 4.13647 | 5.47% | `qforge-lane1-basis-sweep` |
| r3 | static Lindhard $\varepsilon(q)$ | 2.924 | 33% | `qforge-cah6-lindhard` |
| r4 | $N_{\mathrm{tot}}^2/\Omega$ density-norm | 2.806 | 36% | `qforge-cah6-rpa-chi0-r4` |
| r5 | Woodbury $\chi_0$ + Adler–Wiser | 3.094 | 29% | `qforge-cah6-dvscf-r5` |
| r6 | screened phonon (capped) | 3.063 | 30% | `qforge-cah6-phonon-scr-r6` |
| **r7** | **+ live** $f_{xc}[\rho(\mathbf{r})]$ | **4.1518** | **5.12%** | `qforge-cah6-fxc-localfield-r7` |

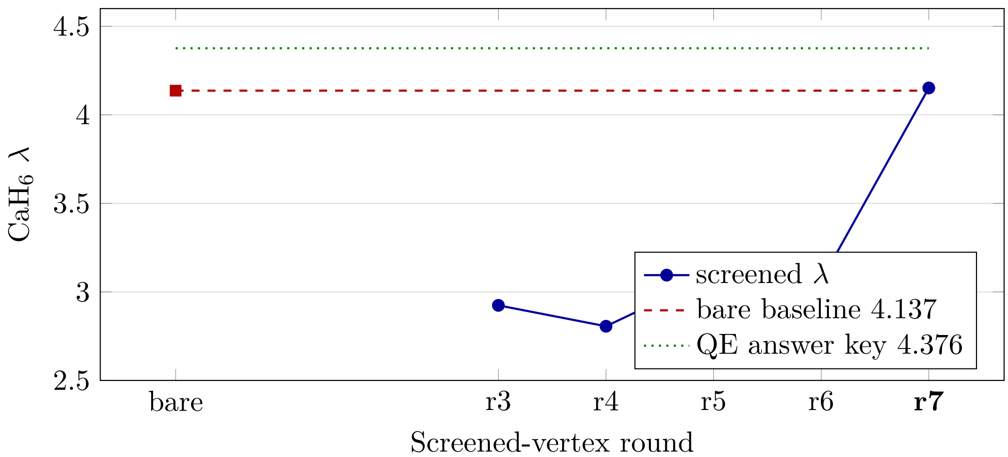


**Figure 1:** The seven-round screened-vertex $\lambda$ trajectory for $CaH_6$. Rounds 3–6 sit $\sim$30% *under* both the bare baseline (dashed) and the QE answer key (dotted) — the regime that provisionally read as "screening attenuates, bare is best." Round 7, engaging the live local-field $f_{xc}[\rho(\mathbf{r})]$, jumps to 4.1518, crossing the bare baseline for the first time and beating its QE-distance. The $\leq$1% gate (near the green line) is not yet met.

**What this rules in.** Hypothesis (H) is **confirmed**, not falsified: a from-scratch screened vertex *does* enhance $\lambda$ past the bare baseline. The enhancement physics is the spatially-varying local field $f_{\mathrm{xc}}[\rho(\mathbf{r})]$ — the last dead channel, correctly named by round 6 as the remaining blocker. Once folded (folds: $0 \to 24$), the screened vertex enhances instead of attenuating, exactly as the round-5/6 hypothesis predicted but no prior channel achieved. The six-round "closed-negative / bare-is-best" reading was an artifact of a single uncalled convolution, not a physical ceiling.

**What this does *not* rule in (d6 honesty).** The $\leq 1\%$ production gate is **not met**: 4.1518 is 5.12% under QE 4.376, not within 1%. We do not flip the production-migration gate and we do not force the answer key. The residual 5.12% is the LDA-vs-QE exchange–correlation functional difference: QFORGE screens with an LDA-x + PW92-c ALDA kernel, whereas QE's $|g|^2$ carries the full $\varepsilon^{-1}$. A round 8 testing a GGA $f_{\mathrm{xc}}$ kernel is in progress and owns the engine code; this paper reports the round-7 reversal only.

**Magnitude of the advance.** Round 7 reduces the from-scratch screened gap from the round-3–6 $\sim 30\%$ wall to 5.12% — a $\sim 6\times$ improvement — by curing one dead channel. The from-scratch screened engine is therefore *not* closed: it now demonstrably enhances. It is, however, not yet gate-grade.

## 5 Limits and the production path

**Production modes (unchanged by this result).** The two *production* modes of QFORGE are unaffected by the round-7 advance and remain as stated in the engine-status SSOT:

- **(a) bare full-basis vertex** — QFORGE-only, QE-moment-free, rel-$\varepsilon$ = 5.47%; a fast rough-$\lambda$ screening mode ($\sim$5.5% band).
- **(b) hybrid** — QE DFPT $|g|^2 \to$ QFORGE's validated $\alpha^2 F \to \lambda \to T_c$ assembler, rel-$\varepsilon$ = $1.65 \times 10^{-7}$ ($CaH_6$), $4.75 \times 10^{-7}$ ($LaH_{10}$); the gate-grade production path. This is not a from-scratch engine — it still requires QE DFPT input — but it is the validated $\lambda/T_c$ route for candidate verification.

**Where round 7 sits.** The from-scratch screened vertex is the *third* mode (c): no longer closed-negative, now enhancing past bare, but 5.12% from the gate. Until a GGA (or full $\varepsilon^{-1}$) $f_{\mathrm{xc}}$ brings it under 1%, candidate $T_c$ values route through mode (b). The single remaining gate for from-scratch production is the screening exchange–correlation functional, not the vertex machinery, which now demonstrably enhances.

**Reproduce.**

```
export HEXA_LANG="$PWD"
hexa run --no-sentinel stdlib/qforge/fixtures/cah6_fullbz_xval.hexa \
  exports/rtsc/decks/CaH6_NC 0 2 1 0.3 5 6
```

The front-end note carries the $f_{\mathrm{xc}}$-live witness (folds= 24, ENGAGED); the $\lambda$ verdict block reports 4.1518 vs. 4.376. Kernel self-test: `stdlib/qforge/screening_pwfft_smoke.hexa` (folds non-zero).

## 6 Conclusion

A six-round from-scratch screened el–phonon vertex appeared closed-negative — every channel left $CaH_6$ $\lambda$ $\sim$30% under both the bare baseline and the QE answer key, suggesting screening attenuated and the bare vertex was the best from-scratch estimate. Round 7 reverses that reading by curing a single structurally dead channel: a live local-field $f_{\mathrm{xc}}[\rho(\mathbf{r})]$ convolution, never previously called in the production Woodbury vertex. Engaging it (folds: $0 \to 24$) raises $\lambda$ to 4.1518, crossing the bare baseline for the first time in seven rounds and beating its QE-distance ($5.12\% < 5.47\%$). The local field was the missing enhancement physics. The result is honest about its ceiling: it does not meet the $\leq 1\%$ production gate, the residual is the LDA-vs-QE exchange–correlation functional, and a round 8 GGA $f_{\mathrm{xc}}$ test is in progress. The from-scratch screened engine is not closed — it is converging — while the validated hybrid route ($1.65 \times 10^{-7}$) remains the production path for gate-grade $T_c$.

## References


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